INCH-POUND
MIL-PRF-81757/16(AS)
w/AMENDMENT 1
26 October 2017
SUPERSEDING
MIL-PRF-81757/16(AS)
1 March 1999

PERFORMANCE SPECIFICATION SHEET

# BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, HIGH RATE, 24-VOLT, 35-AMPERE-HOUR, LOW-MAINTENANCE, VENTED

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81757.

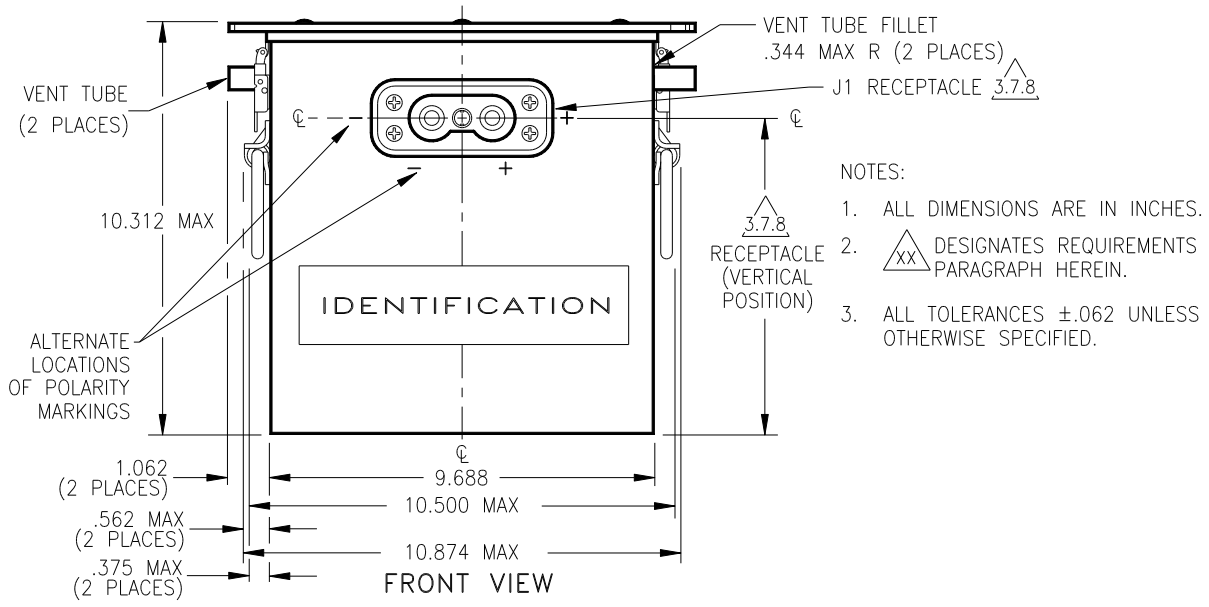



FIGURE 1. Dimensions and configuration for PIN M81757/16-1.

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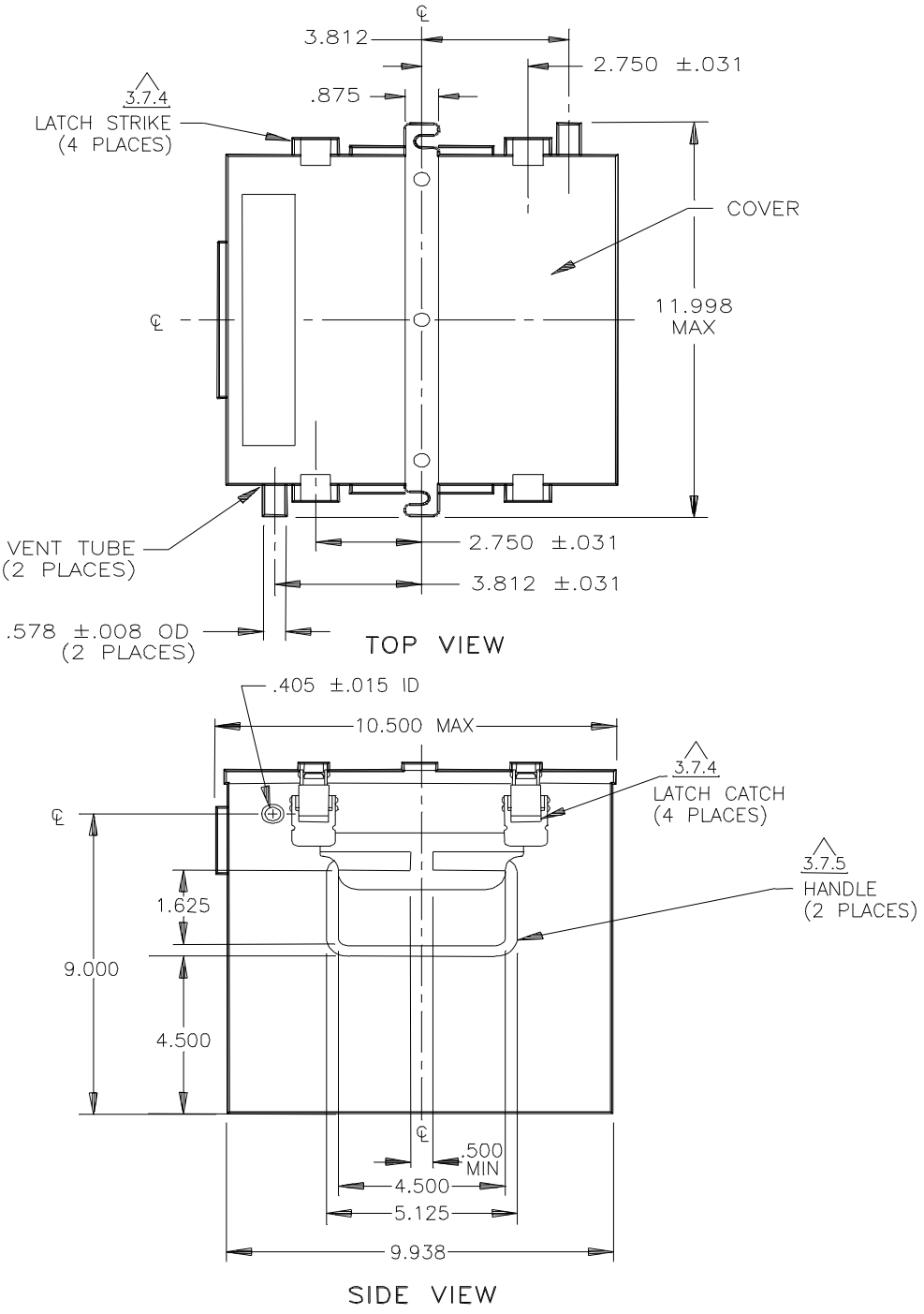
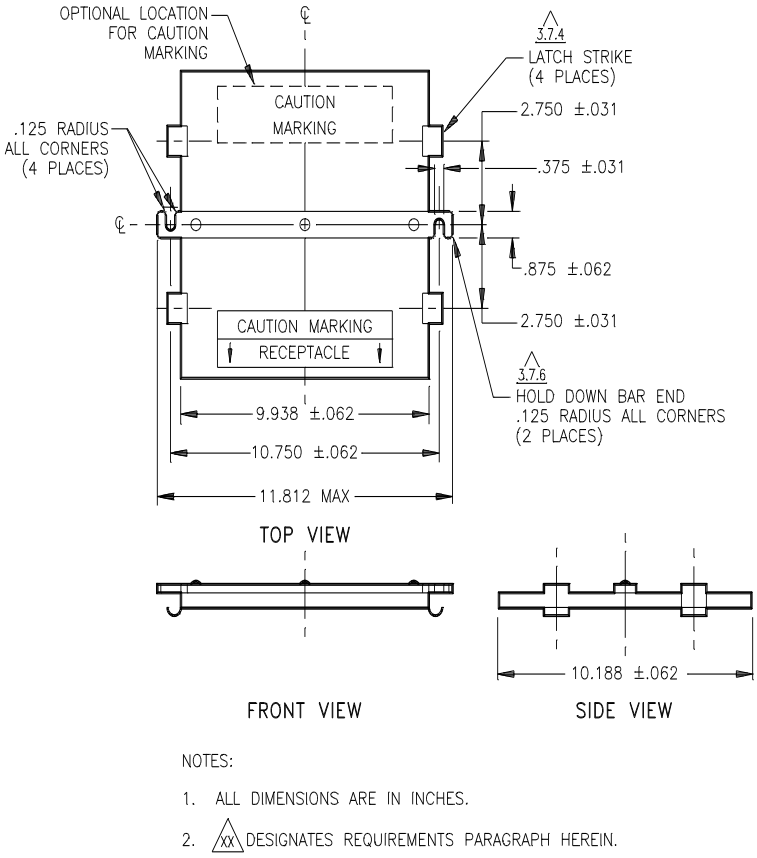



FIGURE 1. Dimensions and configuration for PIN M81757/16-1 - Continued.



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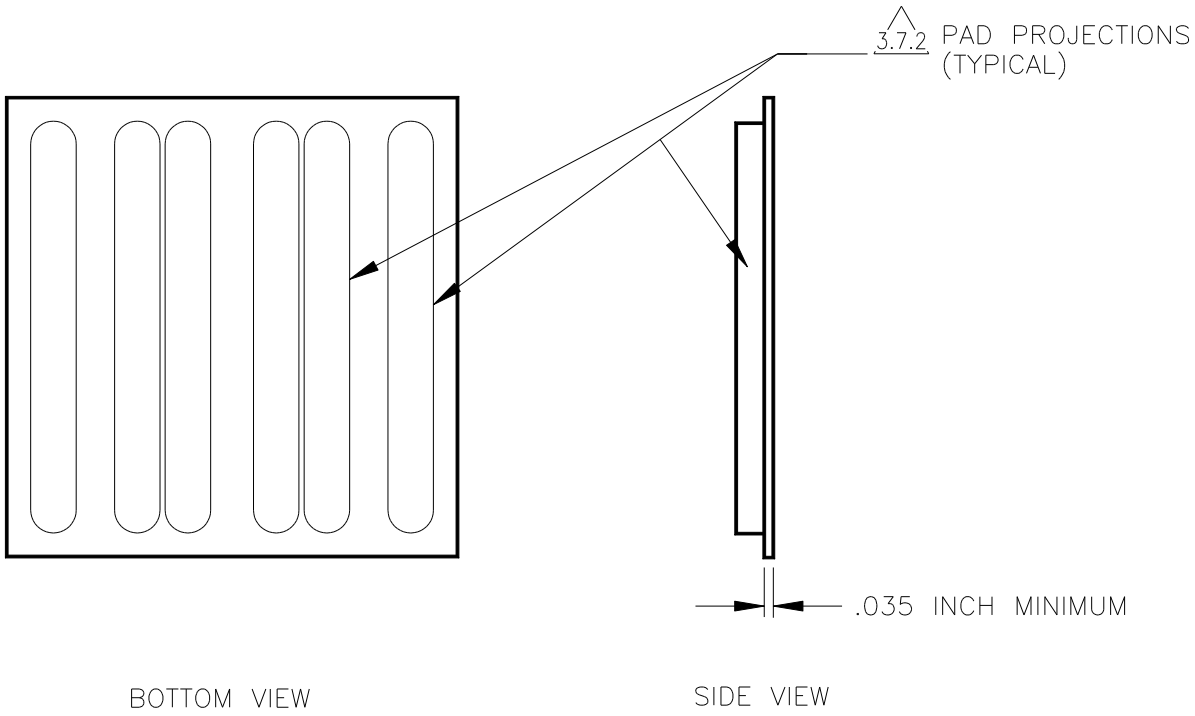

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.

FIGURE 2. Cover for PIN M81757/16-1.



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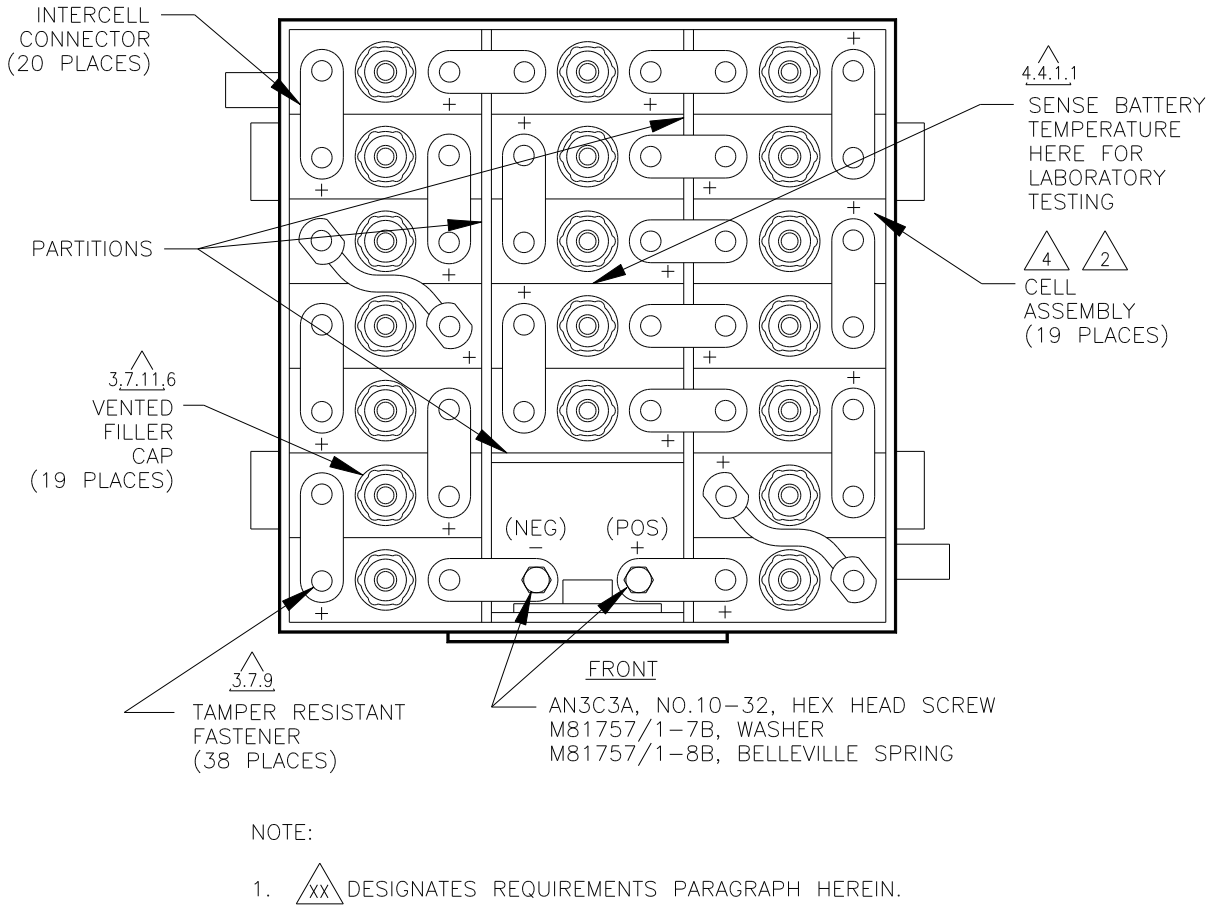

NOTES:

1. PAD SHALL CONFORM TO 3.4, 3.7.2, AND 3.7.3.
2. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.
3. WHEN COVER IS SECURE, THE PROJECTIONS SHALL CONTACT EACH INTERCELL CONNECTOR.

FIGURE 3. Cover gasket/cell hold-down pad for PIN M81757/16-1.



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NOTE:

1. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.

FIGURE 4. Layout of interior components for PIN M81757/16-1.



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| | |
|---|---|
| NOMENCLATURE: | BATTERY, STORAGE, AIRCRAFT NICKEL-CADMIUM, HIGH RATE, LOW-MAINTENANCE, VENTED, LIMITED-REPAIR |
| MILITARY PIN: | M81757/16-1 |
| NATIONAL STOCK NUMBER: | 6140-**-***-**** |
| NOMINAL VOLTAGE: | 24 VOLTS |
| MAXIMUM WEIGHT: | 78 POUNDS |
| INITIAL CAPACITY: | 44 AH (35 AMPS/75 MINUTES/77 °F/19 V) |
| MINIMUM CAPACITY: | 35 AH (35 AMPS/1 HR/77 °F/20 V) |
| CONTRACT AND DELIVERY ORDER NUMBER: | *** |
| MANUFACTURED BY: | *** |
| MANUFACTURER'S CAGE: | *** |
| MANUFACTURER'S SERIAL NUMBER: | *** |
| DATE/LOT CODE: | *** |
| DATE FIRST PLACED IN SERVICE: | |

FIGURE 5. Marking for M81757/16-1 batteries.

| | MINIMUM LETTERING HEIGHT |
|---|---|
| CAUTION | 0.500 inch minimum lettering height. |
| This cover manufactured by CAGE ***** and shall be installed only on a M81757/16-1 battery manufactured by same CAGE. | 0.250 inch minimum lettering height. |
| The cover of other M81757/xx batteries shall not be installed on a M81757/16-1 battery. | 0.250 inch minimum lettering height. |
| The M81757/16-1 battery shall be serviced only in an authorized nickel-cadmium battery shop. The cells shall not be removed. | 0.250 inch minimum lettering height. |
| ⇩ RECEPTACLE ⇩ | 0.500 inch minimum lettering height. |

FIGURE 6. Marking for battery caution and receptacle.



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REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery shall be M81757/16-1. The M81757/16-1 is a limited-repair, low-maintenance, vented battery.

2. Dimensions and weight. The dimensions and weight of the battery shall be as specified in 3.10.8 as modified herein and as shown on figures 1, 2, and 3 contained herein. The exterior dimensions of each cell container shall be identical to the exterior dimensions of PIN M81757/1-6.

3. Capacity. The initial capacity of the battery shall be not less than 44 ampere-hours (Ah). The minimum capacity of the battery shall be not less than 35 Ah. See 3.10.14 as modified herein.

4. Interface requirements between battery and test fixtures. The battery shall contain 19 cells. The cells and other interior components of the battery shall be arranged and positioned as shown on figure 4 contained herein. The cell terminals shall be vertically and horizontally positioned so that each cell terminal shall mate with its matching electrical connection of cell scanner fixture, PIN 371AS156 and cell equalization fixture, PIN 371AS160. The fasteners specified on figure 4 are required due to interfaces with the MS3509-28 per MIL-PRF-18148/3 receptacle, the cell scanner fixture, and the cell equalization fixture.

5. Interface requirements between the battery and the sensor assembly. The battery partition that contains the center row of cells and interfaces with the exterior side wall of cell number 16 (counting from the battery's negative terminal) shall be constructed of thermally conductive material. The height of the partition shall be not less than the height of the plates within cell number 16. The partition's surface adjacent to the side wall of cell number 16 shall be smooth and shall intimately contact the cell's side wall. Spacers or air passages shall not be present at this interface. The partition shall have rigidity and strength to contain cell number 16 for the life of the battery without warping or breaking. The partition shall not distort due to undergoing any or all of the tests herein.

6. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

6.1 Modify the following paragraphs:

3.7.2 Battery container and cover. Delete the entire text and substitute: “The battery container and cover shall be made of a material that is impervious to salt spray and to an aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity (see 6.10.1) and an alternate electrolyte if used. The container and cover may be constructed of materials that are not identical. The cover may be protected with a material (such as paint, epoxy, or powder coating) that is impervious to salt spray and to an aqueous solution of potassium hydroxide of 1.30 ± 0.04



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specific gravity (see 3.8.1 and 6.10.1) and an alternate electrolyte if used. Such protective materials shall not be applied to the battery container except as indicated in 3.8.1. The battery container and cover shall be free of pits, blow holes, rough spots, and other deformations. The dimensions and locations of receptacles, hold-downs, latches (catches and strikes), vent tubes, and battery identification shall conform to figures 1 and 2 of the specification sheet. The cover shall be fitted with a hold-down pad that functions as the cover's gasket and shall have integral projections that hold down each battery cell. The cover gasket/cell hold-down pad shall conform to figure 3 of the specification sheet. The cover pad/gasket shall be sized and fitted to match the rim of the container body for a gastight fit and effective fluid seal between the cover and the container body. The location of the projections shall be determined by the location of each vented filler cap. The projections shall contact each vented filler cap and apply a force on each vented filler cap to restrain vertical movement of the cells during vibration in accordance with 4.5.16. Metal or wood products (such as cork) shall not be used as the gasket. If elastomers are used as the gasket, the elastomer shall conform to the requirements of 3.10.3 (see 6.10.2)."

3.7.8 Receptacles. Following the last sentence, insert: "The J1 receptacle shall conform to MS3509-28. The receptacle shall be vertically positioned so that the top of the terminal block and the cell terminal contact surfaces are in the same plane."

3.7.11.6 Vented filler cap. Following the first sentence, insert: "Each cell shall contain a vented filler cap conforming to PIN M81757/2-7 for the M81757/16-1 battery."

3.8.4.1 Limited-repair batteries. In the first sentence, change "figure 1" to "figure 5 of the specification sheet".

3.8.4.3 Battery caution and receptacle marking. In the second and third sentences, change "figure 3" to "figure 6 of the specification sheet". Insert the following after the third sentence: "The assigned CAGE code shall be inserted at the indicated area."

3.10.8 Dimensions and weights. Delete the entire text and substitute: "The dimensions and configuration of the battery container and container cover shall be as shown on figures 1, 2, and 3 of the specification sheet. The weight of the battery shall be not greater than 78 pounds (35.4 kg) with the electrolyte level adjusted in accordance with 4.4.4."

3.10.14 Capacity. Delete the first sentence and insert: "The initial capacity of the battery shall be not less than 44 Ah. The minimum capacity of the battery shall be not less than 35 Ah."

3.10.14.1 Initial capacity. Delete the entire text and substitute: "To meet the initial capacity requirement, each cell shall produce a potential of not less than 1.0 volt at not less than 75 minutes into the constant 35-ampere discharge after being tested in accordance with 4.5.13 at room temperature (see 4.4.1)."



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3.10.22 Constant voltage discharge. Delete the text and substitute: "The battery shall produce not less than the current specified below after being tested in accordance with 4.5.22."

| Temperature | Average current over the total 60 seconds | Current at 1.0 second | Current at 5.0 seconds | Current at 15 seconds |
|---|---|---|---|---|
| 24.0° ± 2.0°C | 950 A | 1250 A | 1125 A | 1000 A |
| -26.0° ± 2.0°C | 500 A | 600 A | 525 A | 500 A |

3.10.23 Cycling. Delete the text of 3.10.23.c and insert: "c. After successfully completing the required number of duty cycles, the battery's electrical capacity shall comply with the following:

"(1) Qualification inspection: The battery shall deliver not less than 30 Ah when subjected to 4.5.23.d. To comply with the 30-Ah requirement, the battery shall produce a potential of not less than 18.0 volts (end voltage) at not less than 60 minutes into the constant 30-ampere discharge.

"(2) Group C inspection: The battery shall deliver not less than 35 Ah when subjected to 4.5.23.d. To comply with the 35-Ah requirement, the battery shall produce a potential of not less than 20.0 volts (end voltage) at not less than 60 minutes into the constant 35-ampere discharge. The voltage of each cell shall be not less than 1.0 volt at 60 minutes into the discharge."

3.10.23 Cycling. Delete the text of 3.10.23.d and insert: "d. The battery shall successfully complete 700 duty cycles for qualification inspection and 200 duty cycles for group C inspection when subjected to 4.5.23."



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6.2 Add the following paragraph:

"4.4.1.1 Temperature measurement location. When the battery temperature is to be measured during laboratory testing, the temperature shall be measured at the location between the two cells shown on figure 4 of the specification sheet. The temperature shall be measured with a temperature sensor (thermocouple or thermistor) inserted between the two cells and positioned in the approximate center of the outer sidewall of each cell equidistant from top to bottom and side to side."

6.3 Modify the following paragraphs, tables, and figures:

4.5.23 Cycling test. Insert the following after the last sentence: "The battery shall undergo 700 duty cycles for qualification inspection and 200 duty cycles for group C inspection. The temperature near the center of the battery shall be monitored and recorded throughout this test."

4.5.23 Cycling test. Delete the text of 4.5.23.b(1) and insert: "(1) Discharge the battery for 20 seconds at constant current load of 700 amperes."

TABLE III. Qualification inspection of batteries. For the visual and mechanical test (test number 2), add "requirements 4 and 5" to the requirement paragraph column. For the dimensions and weight test (test number 4), add "requirement 2" to the requirement paragraph column. For the initial capacity discharge test (test number 9) and the humidity and charge retention test (test number 10), add "requirement 3" to the requirement paragraph column.

TABLE VII. Group B inspection. Delete the table and insert the table I contained herein.

TABLE VIII. Group C inspection. For the visual and mechanical test (test numbers I-2, II-1, and III-1), add "requirements 4 and 5" to the requirement paragraph column. For the dimensions and weight test (test number I-3), add "requirement 2" to the requirement paragraph column. For the initial capacity discharge tests (test numbers I-6, II-5, and III-4) and the humidity and charge retention test (test number II-6), add "requirement 3" to the requirement paragraph column.

FIGURE 1. Marking for limited-repair batteries. Delete the figure and insert the figure 5 herein.

FIGURE 3. Marking for battery caution and receptacle. Delete the figure and insert the figure 6 herein.



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TABLE I. Group B inspection.

| Test No. | Examinations and Tests | Requirement Paragraph | Test Paragraph | Requirement Paragraph (Spec. Sheet) |
|---|---|---|---|---|
| | Subgroup I: | | | |
| I-1 | Dimensions and weight | 3.10.8, 3.10.9 | 4.5.8 | 2 |
| I-2 | Cover seal test | 3.8.1, 3.9, 3.10.11 | 4.5.10 | |
| I-3 | Vented filler cap test (two vented filler caps per battery) | 3.10.5 | 4.5.5.1 | |
| | Subgroup II: | | | |
| II-1 | Initial capacity discharge | 3.10.14 | 4.5.13 | 3 |
| II-2 | 20-second pulse discharge | 3.9, 3.10.20 | 4.5.20 | |
| | Subgroup III: | | | |
| III-1 | Constant voltage discharge | 3.9, 3.10.22 | 4.5.22 | 6 |
| III-2 | Internal pressure test | 3.9, 3.10.26 | 4.5.26 | |



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Referenced documents. In addition to MIL-PRF-81757, this specification sheet references MIL-PRF-18148/3.

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